



Customer Shopping Behaviour Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions.

Project Overview



The Goal

Analyze transactional data to guide **strategic business decisions**.

Spending Patterns

Customer Segments

Product Preferences

Subscription Behaviour

Dataset at a Glance

3,900

Purchases

18

Columns

37

Missing Values

Review Rating only

Demographics

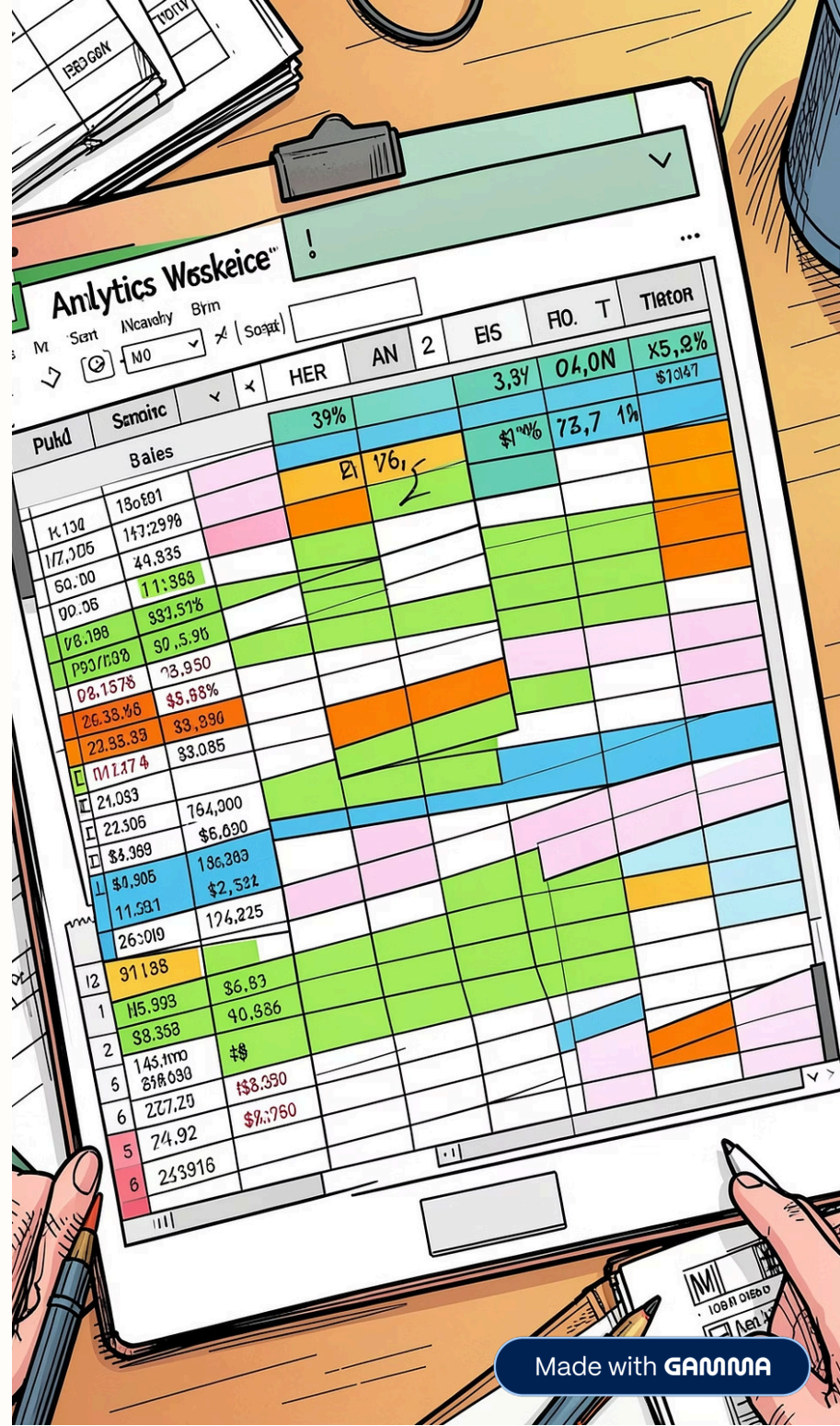
Age, Gender, Location,
Subscription

Purchase Details

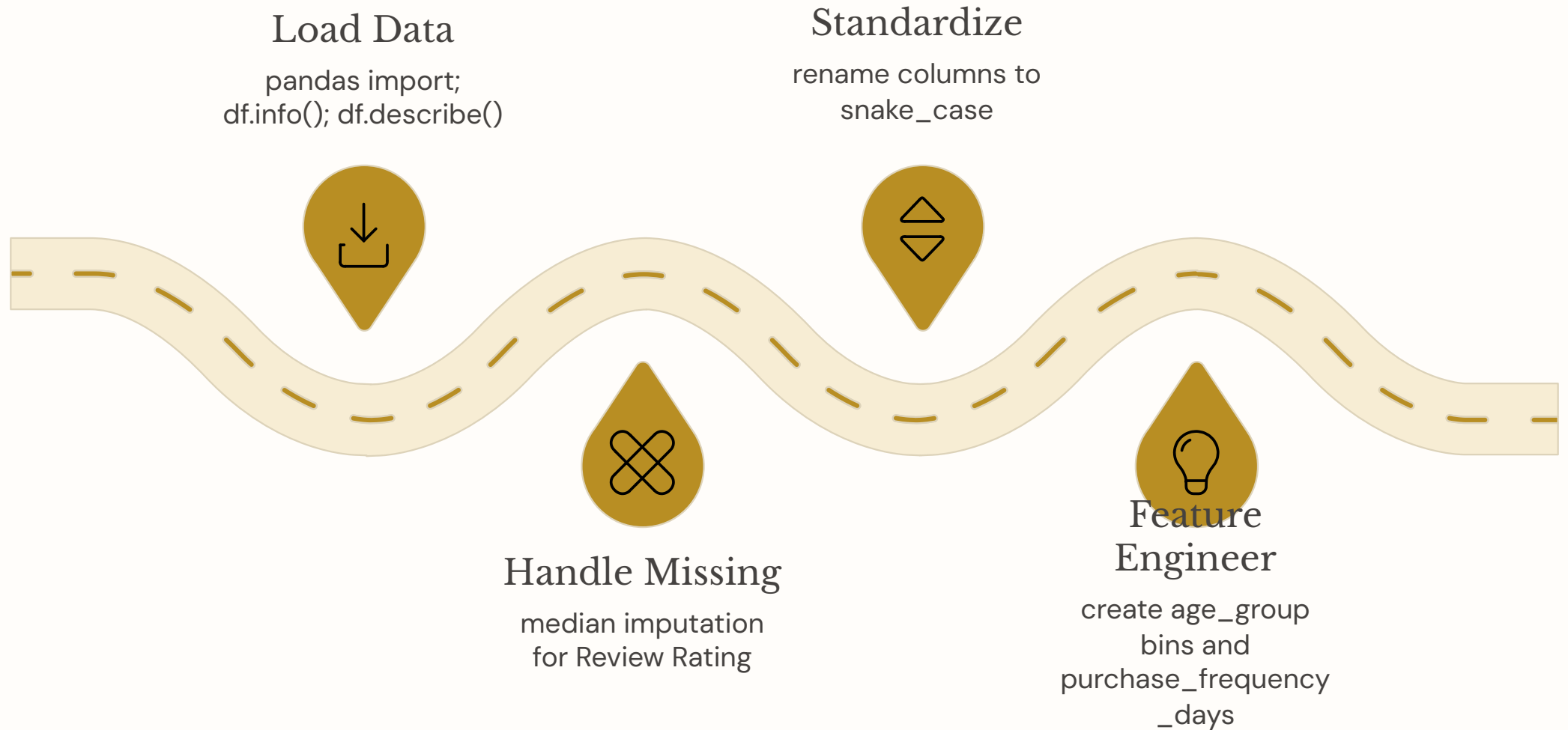
Item, Category, Amount, Season

Behaviour

Discount, Frequency, Rating, Shipping



Data Preparation in Python



Loaded with pandas, explored via `df.info()` and `df.describe()`, then cleaned and engineered before loading into MySQL.

Feature Engineering Highlights

age_group

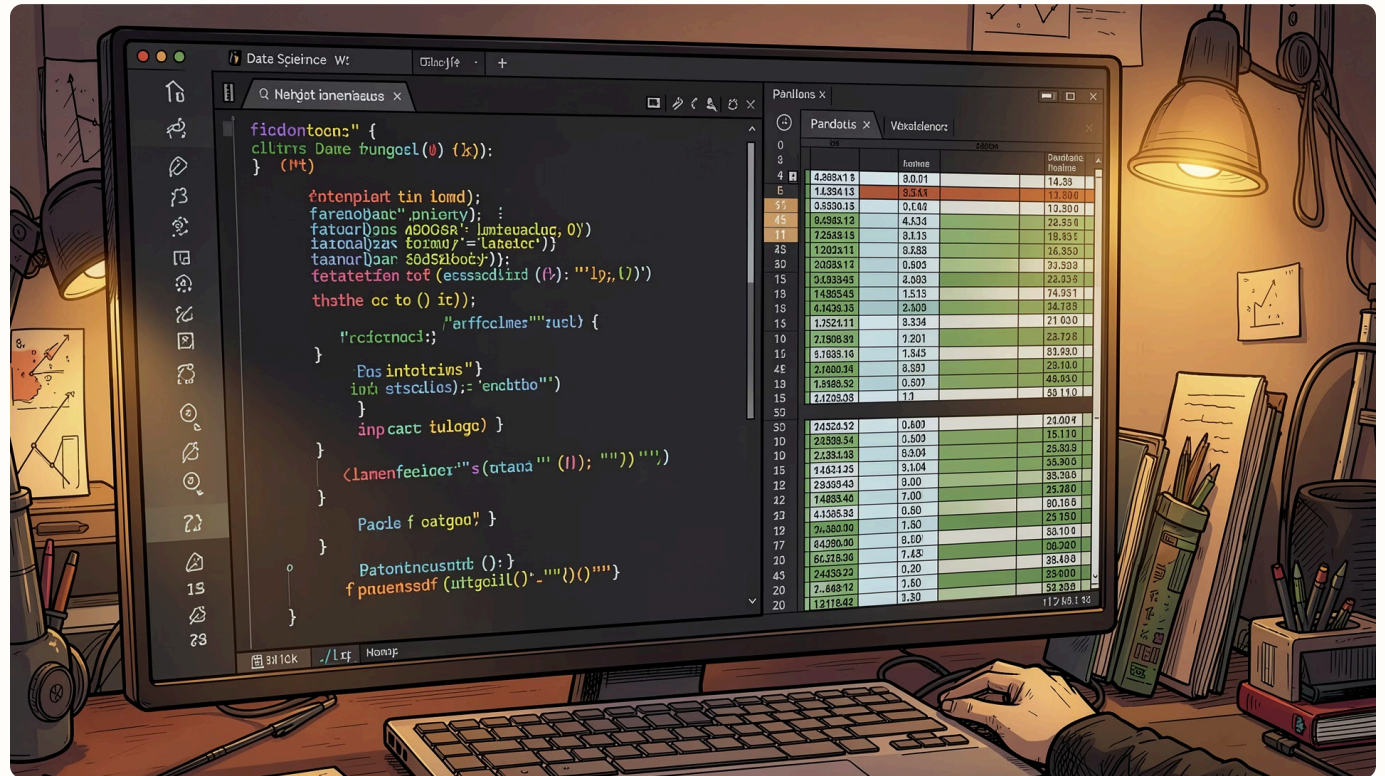
Binned customer ages into segments

purchase_frequency_days

Derived from purchase date data

promo_code_used
dropped

Redundant with discount_applied



Revenue by Gender

GENDER COMPARISON



Key Question

Compared **total revenue** generated by male vs female customers.

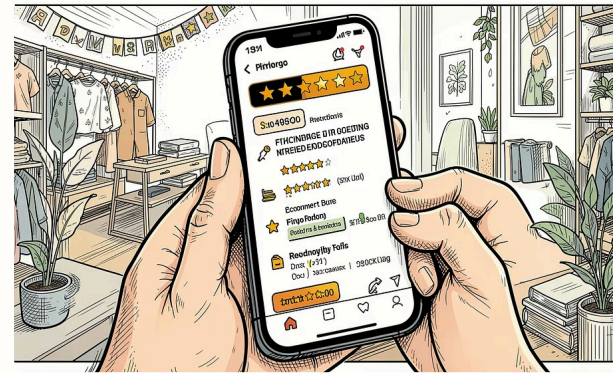
- SQL analysis performed in PostgreSQL to structure business insights.

High-Spenders & Top Products



High-Spending Discount Users

Customers who used discounts but still spent **above average**.



Top 5 Products by Rating

Highest **average review ratings** across all categories.

Shipping & Subscription Insights



Shipping Type Comparison

Average spend: **Standard vs Express** shipping.

Subscribers vs Non-Subscribers

Compared **average spend** and **total revenue** by subscription status.

Discount-Dependent Products

What We Found

Identified the **top 5 products** with the highest percentage of discounted purchases.

- ☐ Reveals which products rely most on promotions to drive sales.



Customer Segmentation



New

First-time purchasers



Returning

Repeat buyers



Loyal

High-frequency, high-value

Customers classified by **purchase history** to enable targeted marketing strategies.

